Chapter 10

Potentials and Fields

Problem 10.1

Show that the differential equations for V and $\boldsymbol{A}$

$$\boldsymbol{\nabla}^2 V + \frac{\partial}{\partial t}(\boldsymbol{\nabla}\cdot\boldsymbol{A}) = -\frac{1}{\epsilon_0}\rho$$

$$\left(\boldsymbol{\nabla}^2\boldsymbol{A} - \mu_0\epsilon_0\frac{\partial^2\boldsymbol{A}}{\partial t^2}\right) - \boldsymbol{\nabla}\left(\boldsymbol{\nabla}\cdot\boldsymbol{A} + \mu_0\epsilon_0\frac{\partial V}{\partial t}\right) = -\mu_0\boldsymbol{J}$$

can be written in the more symmetrical form

$$\Box^2 V + \frac{\partial L}{\partial t} = -\frac{1}{\epsilon_0}\rho$$

$$\Box^2\boldsymbol{A} - \boldsymbol{\nabla}L = -\mu_0\boldsymbol{J}$$

where

$$\Box^2 \equiv \boldsymbol{\nabla}^2 - \mu_0\epsilon_0\frac{\partial^2}{\partial t^2} \quad \text{and} \quad L \equiv \boldsymbol{\nabla}\cdot\boldsymbol{A} + \mu_0\epsilon_0\frac{\partial V}{\partial t}$$

Solution:

Put the definitions of $\Box^2$ and L into the equations for V:

$$\left(\boldsymbol{\nabla}^2 - \mu_0\epsilon_0\frac{\partial^2}{\partial t^2}\right)V + \frac{\partial}{\partial t}\left(\boldsymbol{\nabla}\cdot\boldsymbol{A} + \mu_0\epsilon_0\frac{\partial V}{\partial t}\right) =$$

$$= \boldsymbol{\nabla}^2 V - \mu_0\epsilon_0\frac{\partial^2 V}{\partial t^2} + \frac{\partial}{\partial t}(\boldsymbol{\nabla}\cdot\boldsymbol{A}) + \mu_0\epsilon_0\frac{\partial^2 V}{\partial t^2} =$$

$$= \boldsymbol{\nabla}^2 V + \frac{\partial}{\partial t}(\boldsymbol{\nabla}\cdot\boldsymbol{A}) = -\frac{1}{\epsilon_0}\rho$$

and for $\boldsymbol{A}$:

$$\left(\boldsymbol{\nabla}^2 - \mu_0\epsilon_0\frac{\partial^2}{\partial t^2}\right)\boldsymbol{A} - \boldsymbol{\nabla}\left(\boldsymbol{\nabla}\cdot\boldsymbol{A} + \mu_0\epsilon_0\frac{\partial V}{\partial t}\right) =$$

$$= \left(\boldsymbol{\nabla}^2\boldsymbol{A} - \mu_0\epsilon_0\frac{\partial^2\boldsymbol{A}}{\partial t^2}\right) - \boldsymbol{\nabla}\left(\boldsymbol{\nabla}\cdot\boldsymbol{A} + \mu_0\epsilon_0\frac{\partial V}{\partial t}\right) = -\mu_0\boldsymbol{J}$$

□

Problem 10.2

Problem 10.3

(a) Find the fields, and the charge and current distributions, corresponding to

$$V(\boldsymbol{r},t) = 0, \quad \boldsymbol{A}(\boldsymbol{r},t) = -\frac{1}{4\pi\epsilon_0}\frac{qt}{r^2}\hat{\mathbf{r}}$$

(b) Use the gauge function $\lambda = -(1/4\pi\epsilon_0)(qt/r)$ to transform the potentials, and comment on the result.

Solution:

The **E** and **B** fields are

$$\mathbf{E} = -\boldsymbol{\nabla}V - \frac{\partial\mathbf{A}}{\partial t}$$
$$\mathbf{B} = \boldsymbol{\nabla}\times\mathbf{A}$$

The derivative of **A** with respect to time is

$$\frac{\partial\mathbf{A}}{\partial t} = -\frac{1}{4\pi\epsilon_0}\frac{q}{r^2}\hat{\mathbf{r}},$$

while the curl of **A** in spherical coordinates is

$$\frac{1}{r\sin\theta}\left[\frac{\partial}{\partial\theta}(\sin\theta A_\phi) - \frac{\partial A_\theta}{\partial\phi}\right]\hat{\mathbf{r}} + \frac{1}{r}\left[\frac{1}{\sin\theta}\frac{\partial A_r}{\partial\phi} - \frac{\partial}{\partial r}(rv_\phi)\right]\hat{\boldsymbol{\theta}} + \frac{1}{r}\left[\frac{\partial}{\partial r}(rA_\theta) - \frac{\partial A_r}{\partial\theta}\right]\hat{\boldsymbol{\phi}}.$$

Since $A_\phi = A_\theta = 0$, and $\frac{\partial A_r}{\partial\phi} = \frac{\partial A_r}{\partial\theta} = 0$, the curl of **A** is zero.

(a)

$$\mathbf{E} = -\frac{\partial \mathbf{A}}{\partial t} = \frac{1}{4\pi\epsilon_0}\frac{q}{r^2}\hat{\mathbf{r}}$$
$$\mathbf{B} = \boldsymbol{\nabla} \times \mathbf{A} = 0$$

(b) Using the gauge function, V and $\mathbf{A}$ are transformed into

$$V'(\mathbf{r}, t) = V(\mathbf{r}, t) - \frac{\partial \lambda}{\partial t}$$
$$\mathbf{A}'(\mathbf{r}, t) = \mathbf{A}(\mathbf{r}, t) + \boldsymbol{\nabla}\lambda$$

Writing out the derivative and gradient of λ

$$\frac{\partial \lambda}{\partial t} = -\frac{1}{4\pi\epsilon_0}\frac{q}{r}$$
$$\boldsymbol{\nabla}\lambda = \frac{1}{4\pi\epsilon_0}\frac{qt}{r^2}\hat{\mathbf{r}}$$

gives the transformed potentials

$$V'(\mathbf{r}, t) = \frac{1}{4\pi\epsilon_0}\frac{q}{r}$$
$$\mathbf{A}'(\mathbf{r}, t) = -\frac{1}{4\pi\epsilon_0}\frac{qt}{r^2}\hat{\mathbf{r}} + \frac{1}{4\pi\epsilon_0}\frac{qt}{r^2} = 0$$

$\mathbf{E}$ is now given by $-\boldsymbol{\nabla}V$ (instead of by $-\frac{\partial \mathbf{A}}{\partial t}$), while $\mathbf{B}$ is again 0, this time directly because $\boldsymbol{\nabla} \times \mathbf{A} = \boldsymbol{\nabla} \times 0 = 0$.

$$\mathbf{E} = -\boldsymbol{\nabla}V = -\frac{1}{4\pi\epsilon_0}\frac{q}{r^2}\hat{\mathbf{r}}$$
$$\mathbf{B} = \boldsymbol{\nabla} \times \mathbf{A} = 0$$

The result is exactly the same as in (a).

□

Problem 10.4

Suppose $V = 0$ and $\mathbf{A} = A_0 \sin(kx - \omega t)\hat{\mathbf{y}}$, where $A_0, \omega,$ and k are constants. Find $\mathbf{E}$ and $\mathbf{B}$, and check that they satisfy Maxwell's equations in vacuum. What conditions must you impose on ω and k?

Solution:

Maxwell's equations in vacuum are

$$\text{(i)}\nabla \cdot \mathbf{E} = \frac{1}{\epsilon_0}\rho \qquad \text{(iii)}\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

$$\text{(ii)}\nabla \cdot \mathbf{B} = 0 \qquad \text{(iv)}\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}$$

where as before we set

$$\mathbf{E} = -\nabla V - \frac{\partial \mathbf{A}}{\partial t}$$
$$\mathbf{B} = \nabla \times \mathbf{A}$$

This gives

$$\mathbf{E} = -\frac{\partial \mathbf{A}}{\partial t} = -\omega A_0 \cos(kx - \omega t)\hat{\mathbf{y}}$$
$$\mathbf{B} = \frac{\partial A_y}{\partial x}\hat{\mathbf{z}} = kA_0 \cos(kx - \omega t)\hat{\mathbf{z}}$$

Since $\rho = 0$, this directly gives $\nabla \cdot \mathbf{E} = 0$. (ii) gives $\nabla \cdot \mathbf{B} = 0$.

From (iv) we have the relation, since the current density $\mathbf{J} = 0$,

$$\nabla \times \mathbf{B} = \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}$$
$$\downarrow$$
$$\nabla \times \mathbf{B} = -\frac{\partial B_z}{\partial x}\hat{\mathbf{y}} = k^2 A_0 \sin(kx - \omega t)\hat{\mathbf{y}}$$
$$\mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} = \mu_0 \epsilon_0 \omega^2 A_0 \sin(kx - \omega t)\hat{\mathbf{y}}$$

From this we see that $k^2 = \mu_0 \epsilon_0 \omega^2 \rightarrow k = \sqrt{\mu_0 \epsilon_0}\omega \rightarrow k = \frac{\omega}{c}$

From (iii) we have the relation

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$
$$\downarrow$$
$$\nabla \times \mathbf{E} = \frac{\partial E_y}{\partial x}\hat{\mathbf{z}} = \omega k A_0 \sin(kx - \omega t)\hat{\mathbf{z}}$$
$$-\frac{\partial \mathbf{B}}{\partial t} = \omega k A_0 \sin(kx - \omega t)\hat{\mathbf{z}}$$

Equations (i)-(iv) are all verified with the condition $k = \frac{\omega}{c}$.

□

Problem 10.5

Which of the potentials

$$\text{(i) } V = 0, \quad \mathbf{A} = \begin{cases} \frac{\mu_0 k}{4c}(ct - |x|)^2 \hat{\mathbf{z}}, & \text{for } |x| < ct \\ 0 & \text{for } |x| > ct \end{cases}$$

$$\text{(ii) } V = 0, \quad \mathbf{A} = -\frac{1}{4\pi\epsilon_0}\frac{qt}{r^2}\hat{\mathbf{r}}$$

$$\text{(iii) } V = 0, \quad \mathbf{A} = A_0 \sin(kx - \omega t)\hat{\mathbf{y}}$$

are in the Coulomb gauge? Which are in the Lorenz gauge?

Solution:

The Coulomb gauge is

$$\boldsymbol{\nabla} \cdot \mathbf{A} = 0$$

and the Lorenz gauge is

$$\boldsymbol{\nabla} \cdot \mathbf{A} = -\mu_0 \epsilon_0 \frac{\partial V}{\partial t}$$

Starting with checking the Coulomb gauge, we get

$$\text{(i) } \boldsymbol{\nabla} \cdot \mathbf{A} = \boldsymbol{\nabla} \cdot \left(\frac{\mu_0 k}{4c}(ct - |x|)^2 \hat{\mathbf{z}} \right) = 0$$

$$\text{(ii) } \boldsymbol{\nabla} \cdot \mathbf{A} = \boldsymbol{\nabla} \cdot \left(-\frac{1}{4\pi\epsilon_0}\frac{qt}{r^2}\hat{\mathbf{r}} \right) = -\frac{qt}{4\pi\epsilon_0} \boldsymbol{\nabla} \cdot \left(\frac{\hat{\mathbf{r}}}{r^2} \right) = -\frac{qt}{4\pi\epsilon_0}\delta^3(\mathbf{r})$$

$$\text{(iii) } \boldsymbol{\nabla} \cdot \mathbf{A} = \boldsymbol{\nabla} \cdot (A_0 \sin(kx - \omega t)\hat{\mathbf{y}}) = 0$$

(i) and (iii) are in the Coulomb gauge, but not (ii). Next, checking Lorenz gauge we see that the divergence of $\mathbf{A}$ should be proportional to the time derivative of V, but since $V = 0$ for (i) and (iii), they are also in the Lorenz gauge. This does not apply to (ii) since the divergence of $\mathbf{A} \neq 0$ while $\frac{\partial V}{\partial t} = 0$.

(i) Both Coulomb and Lorenz.

(ii) Neither Coulomb or Lorenz.

(iii) Both Coulomb and Lorenz. □

Problem 10.6

Problem 10.7

Problem 10.8

Problem 10.9

Problem 10.10

Problem 10.11

Problem 10.12

Problem 10.13

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